

Although estimators such as d_n , called *superefficient*, can be constructed in some generality, they are more of a theoretical oddity than a practical concern. This is because the values of θ for which the variance goes below the bound are a set of Lebesgue measure 0. However, the existence of superefficient estimators serves to remind us to always be careful in our examination of assumptions for establishing properties of estimators (and to be careful in general!).

10.6.2 Suitable Regularity Conditions

The phrase “under suitable regularity conditions” is a somewhat abused phrase, as with enough assumptions we can probably prove whatever we want. However, “regularity conditions” are typically very technical, rather boring, and usually satisfied in most reasonable problems. But they are a necessary evil, so we should deal with them. To be complete, we present a set of regularity conditions that suffice to rigorously establish Theorems 10.1.6 and 10.1.12. These are not the most general conditions but are sufficiently general for many applications (with a notable exception being if the MLE is on the boundary of the parameter space). Be forewarned, the following is not for the fainthearted and can be skipped without sacrificing much in the way of understanding.

These conditions mainly relate to differentiability of the density and the ability to interchange differentiation and integration (as in the conditions for Theorem 7.3.9). For more details and generality, see Stuart, Ord, and Arnold (1999, Chapter 18), Ferguson (1996, Part 4), or Lehmann and Casella (1998, Section 6.3).

The following four assumptions are sufficient to prove Theorem 10.1.6, consistency of MLEs:

- (A1) We observe $X_1, \dots, X_n$, where $X_i \sim f(x|\theta)$ are iid.
- (A2) The parameter is *identifiable*; that is, if $\theta \neq \theta'$, then $f(x|\theta) \neq f(x|\theta')$.
- (A3) The densities $f(x|\theta)$ have common support, and $f(x|\theta)$ is differentiable in θ .
- (A4) The parameter space Ω contains an open set ω of which the true parameter value θ_0 is an interior point.

The next two assumptions, together with (A1)–(A4) are sufficient to prove Theorem 10.1.12, asymptotic normality and efficiency of MLEs.

- (A5) For every $x \in \mathcal{X}$, the density $f(x|\theta)$ is three times differentiable with respect to θ , the third derivative is continuous in θ , and $\int f(x|\theta) dx$ can be differentiated three times under the integral sign.
- (A6) For any $\theta_0 \in \Omega$, there exists a positive number c and a function $M(x)$ (both of which may depend on θ_0) such that

$$\left| \frac{\partial^3}{\partial \theta^3} \log f(x|\theta) \right| \leq M(x) \quad \text{for all } x \in \mathcal{X}, \quad \theta_0 - c < \theta < \theta_0 + c.$$

with $E_{\theta_0}[M(X)] < \infty$.